Valence Bond Theory

VBT.

Assignment Title

"Hybridization"

Course:

FUNDAMENTAL CONCEPT OF CHEMICAL BONDING

Hybridization of Orbitals

The concept of hybridization was introduced by **Linus Pauling**, an American chemist, in **1930s**.

Definition:

It can be defined as:

"The mixing of atomic orbitals to give equal number of symmetrically hybridized orbitals is reffered to as hybridization."

or

"The phenomenon of mixing up of atomic orbitals of similar energies and formation of equivalent number of entirely new orbitals of identical shape and energy is known as hybridization and the new orbitals so formed is called as hybrid orbitals."

Types of hybridization:

There are eight types of hybridization are following:

(i) sp - hybridization

(ii) sp^2 - hybridization

(iii) sp^3 - hybridization

(iv) dsp^3 - hybridization

(v) d^2sp^3 - hybridization

(vi) d^3sp^3 - hybridization

(vii) d^4sp^3 - hybridization

(viii) d^5sp^3 - hybridization

Rules for Hybridization:

Some of the rules for orbitals hybridization are as follows:

i- Number of hybrid orbitals produced, is equal to the number of orbitals undergoing hybridization.

ii- The type of hybridization can predict the geometry and bond angles of a molecule.

iii- The orbital which has been used to build up a hybrid orbital becomes no longer available to hold electron in its pure form.

iv- Orbitals of similar energies belonging to the same atom or ion can hybridize together.

Coordination Number	Shape of molecules	Scheme of hybridiza
02	Linear	pd, sd
03	Trigonal planar	p^2d
	Unsymmetrical	spd
	Trigonal pyramidal	pd^2
04	Tetrahedral	sd^3
	Irregular tetrahedral	spd^2, p^3d, pd^3
	Square planer	p^2d^2
05	Trigonal bipyramidal	spd^3
	pentagonal planer	p^2d^3
	Tetragonal pyramidal	sp^2d^2, sd^4, pd^4, p
06	Trigonal prism	spd^4, pd^5
	Trigonal antiprism	p^3d^3
07	Capped trigonal prism	d^4sp^2
08	Dodecahedral	d^4sp^3
	Square antiprismatic	d^5sp^3
09	Tricapped trigonal prismatic	d^5sp^3

Sp-hybridization

"The combination of one **s** and one **p** orbitals to form two hybrid orbitals of same shape and equal energy is called as **sp-hybridization**."

Examples: AX$_2$ Type Compounds

i- BeF$_2$

Central Atom: Be

$_4Be = 1s^2 \cdot 2s^{\uparrow\downarrow} \cdot 2p_x^0 \cdot 2p_y^0 \cdot 2p_z^0$

2s [↑↓] 2p [| |] (Ground state)

2s [↑] 2px [↑] 2py [ ] 2pz [ ] (Excited state)

Hybridization: sp

sp [↑] sp [↑]
F F

(hybridized state)

Bond pair: 02 **Lone Pair:** 0 **Steric no:** 2+0=2

Orbital Diagram:

* 50% s-character
* 50% p-character

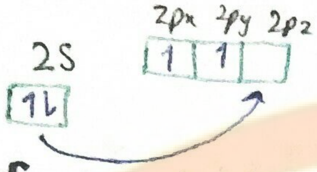
Electronic geometry / Molecular Shape: Linear F —— Be —— F

Bond Angle: 180°

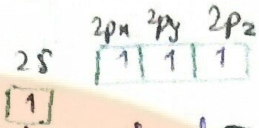
ii- CO_2

Central atom: Carbon

$_6C = 1s^2 \cdot 2s^{\uparrow\downarrow} \cdot 2p_x^1 \cdot 2p_y^1 \cdot 2p_z^0$

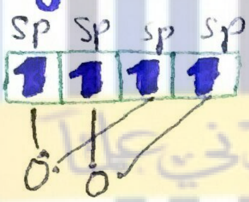


{Ground state}



{Excited state}

Hybridization: sp



{Hybridized state}

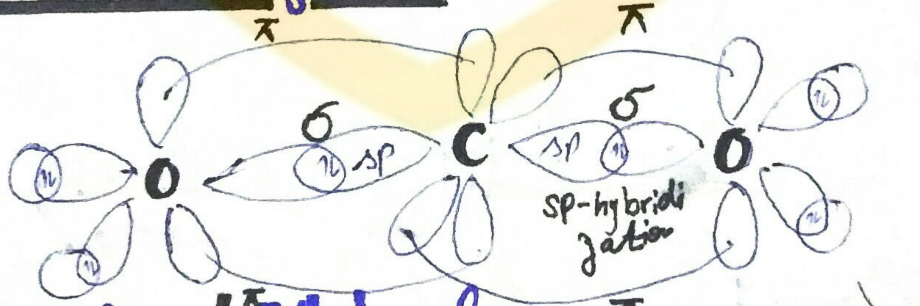
Lone Pair: 0

Bond pair: 2

Steric no: 0+2=2

Bond Angle: 180°

Orbital diagram:



Electronic / Molecular geometry / Shape: Linear

$:\ddot{O}=C=\ddot{O}:$

iii - CS_2

Central Atom: Carbon

$_6C = 1s^2 . 2s^2 . 2p_x^1 . 2p_y^1 . 2p_z^0$

2s	2px	2py	2pz
1↓	1	1	

{Ground state}

2s	2px	2py	2z
1	1	1	1

{Excited state}

sp	sp	sp	sp
1	1	1	1

S S

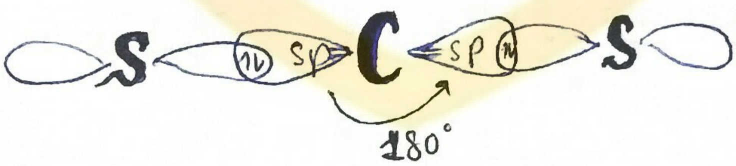
{Hybridized state}

Hybridization: sp

Steric No: 0+2=2 **Lone pair:** 0

Bond pair: 02 **Bond angle:** 180°

Orbital Diagram:



Electronic Geometry / **Molecular Shape**: Linear

$S = C = S$

iv- $HgCl_2$

Central Atom: Mercury

$_{80}Hg = 1s^2 \cdot 2s^2 \cdot 2p^6 \cdot 3s^2 \cdot 3p^6 \cdot 4s^2 \cdot 3d^{10} \cdot 4p^6 \cdot 5s^2 \cdot 4d^{10} \cdot 5p^6 \cdot 6s^2 \cdot 4f^{14} \cdot 5d^{10}$

6s [1↓] 6px [ ] 6py [ ] 6pz [ ] {Ground state}

6s [1] 6px [1] 6py [ ] 6pz [ ] {Excited state}

sp [1] sp [1] {Hybridized state}

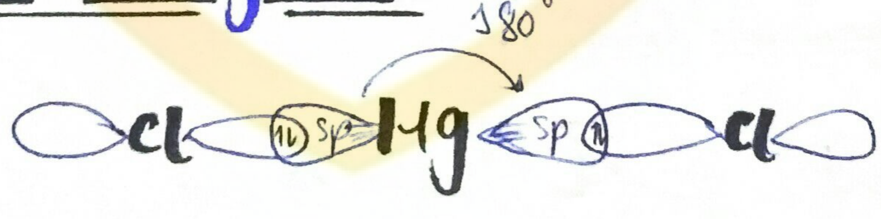
Cl Cl

Hybridization: sp

Steric no: 0+2 = 2 **Lone pair:** 0

Bond pair: 2 **Bond angle:** 180°

Orbital Diagram:



Electronic / Molecular geometry / Shape: Linear

$:\ddot{\underset{..}{Cl}} - Hg - \ddot{\underset{..}{Cl}}:$

V- C_2H_2

Central Atom: Carbon

$_6C = 1s^2 \cdot 2s^2 \cdot 2p_x^1 \cdot 2p_y^1 \cdot 2p_z^0$

2s [↑↓] 2p [↑][↑][ ] {Ground state}

2s [↑] $2p_x$ $2p_y$ $2p_z$ [↑][↑][↑] {Excited state}

↓

sp sp $2p_y^1$ $2p_z^1$ [↑][↑] [↑][↑] {Hybridized state}

H H

Hybridization: sp

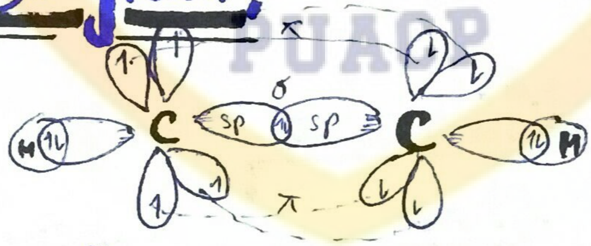
Steric No: 0+2=0

Bond pair: 02

Lone pair: 0

Bond Angle: 180°

Orbital Diagram:



Electronic Geometry / Molecular Shape: Linear

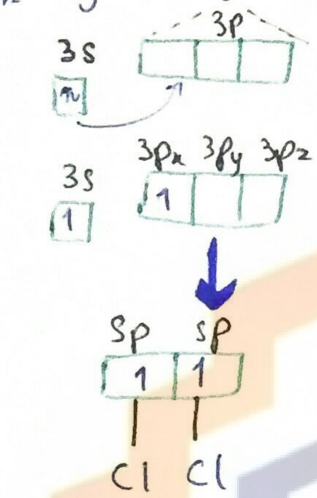
$$H - C \equiv C - H$$

180°

vi- $MgCl_2$

Central Atom: Magnesium

$_{12}Mg = 1s^2 \cdot 2s^2 \cdot 2p^6 \cdot 3s^2 \cdot 3p^0$



{Ground state}

{Excited state}

{Hybridized state}

Hybridization: sp

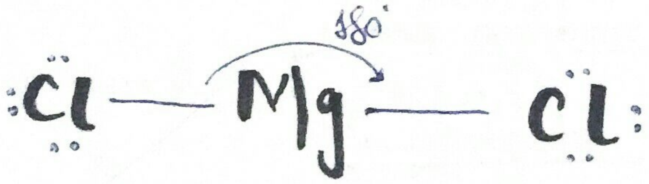
Lone pair: 0

Steric No: 0+2 = 2

Bond pair: [illegible]

Orbital Diagram:

Electronic Geometry / Molecular Shape: Linear



Bond Angle: 180°

sp^2- hybridization AX_3

"The combination of one s and two p orbitals to from three hybrid orbitals of same shape and equal energy is called as the **sp^2-hybridization.**"

Examples:

AX_3 Type Compounds

The three electron groups around central atom repel each other. They lie at corners of an equilateral triangle. This is the trigonal planer arrangement, and the bond angle is **120°**.

i- BF_3

Central Atom: Boron

$_5B: 1s^2 \cdot 2s^2 \cdot 2p_x^1 \cdot 2p_y^0 \cdot 2p_z^0$

2s [1↓] 2p [1][ ][ ] {Ground state}

2s [1] $2p_x$ $2p_y$ $2p_z$ [1][1][ ] {Excited state}

sp^2 sp^2 sp^2 [1][1][1] {Hybridized state}

F F F

It has **33.3%** s-character and **66.6%** p-charater.

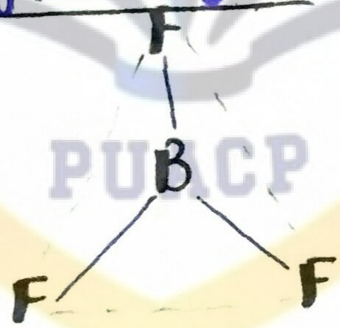
Hybridization: sp^2

Steric No: $0+3=3$ **Lone pair:** 0

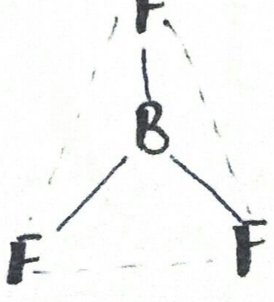
Bond pair: 3 **Bond Angle:** 120°

Orbital Diagram:

Electronic Geometry: Trigonal planer



Molecular Shape: Trigonal planer



ii - BCl_3

Central Atom: Boron

$_5B = 1s^2 \cdot 2s^2 \cdot 2p_x^1 \cdot 2p_y^0 \cdot 2p_z^0$

2s: [1↓] 2px 2py 2pz: [1 | |] {Ground State}

2s: [1] 2px 2py 2pz: [1 | 1 |] {Excited state}

sp^2 sp^2 sp^2: [1 | 1 | 1] Cl Cl Cl {Hybridized state}

Hybridization: sp^2

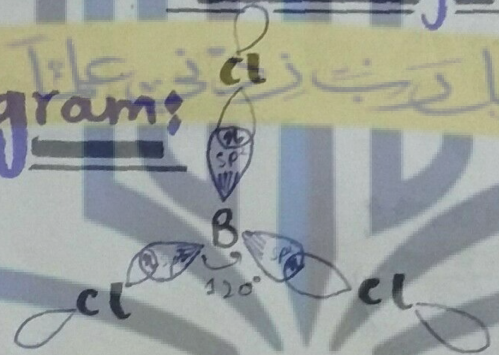
Steric No: $0+3=3$

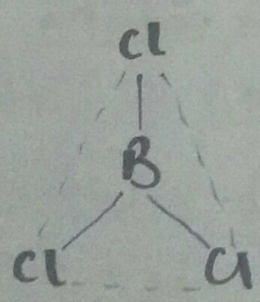
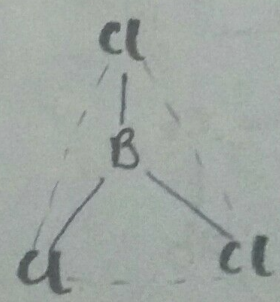
Bond pair: 3

Lone pair: 0

Bond angle: 120°

Orbital Diagram:



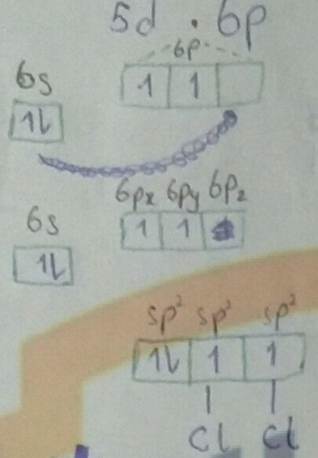
Electronic geometry	**Molecular shape**
Trigonal planer	Trigonal planer
	

AX$_2$E

iii- $PbCl_2$

Central Atom: Lead

$_{82}Pb = 1s^2 \cdot 2s^2 \cdot 2p^6 \cdot 3s^2 \cdot 3p^6 \cdot 4s^2 \cdot 3d^{10} \cdot 4p^6 \cdot 5s^2 \cdot 4d^{10} \cdot 5p^6 \cdot 6s^2 \cdot$
$5d^{10} \cdot 6p^2$



{Ground state}

{Excited state}

{Hybridized state}

Hybridization: sp^2

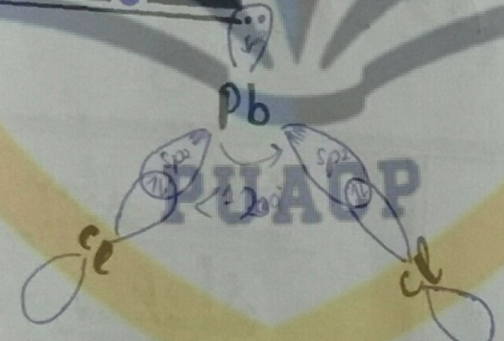
Steric No: $1 + 2 = 3$

Bond pair: 2

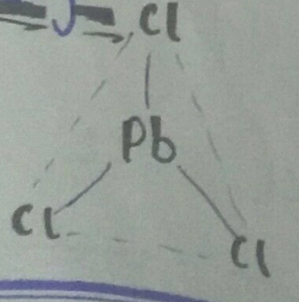
Lone pair: 01

Bond Angle: $< 120°$

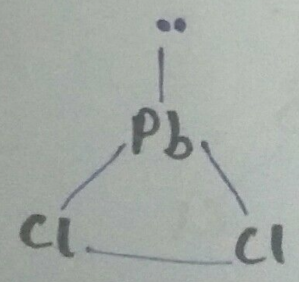
Orbital diagram:



Electronic Geometry Trigonal planer



Molecular Shape Angul... V-sha... Ben...



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vi- Cl_2O

Central Atom: Oxygen

$_8O = 1s^2 \cdot 2s^2 \cdot 2p_x^{\uparrow\downarrow} \cdot 2p_y^{1} \cdot 2p_z^{1}$

2s [↑↓] 2px [↑↓] 2py [↑] 2pz [↑] Ground state

sp³ [↑↓] sp³ [↑↓] sp³ [↑] sp³ [↑] Hybridized state

Cl Cl

Hybridization: sp^3

Steric No: $2+2=4$

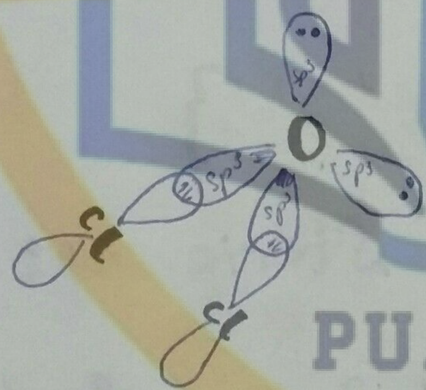
Lone pair: 02

Bond pair: 02

Bond Angle: $< 109.5°$

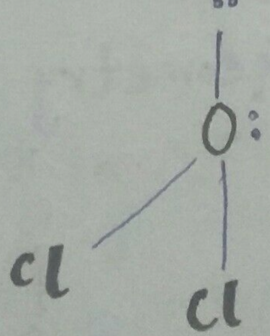
Orbital Diagram:

Tetrahedral



Molecular shape:

V-Shape / Bent / Angular



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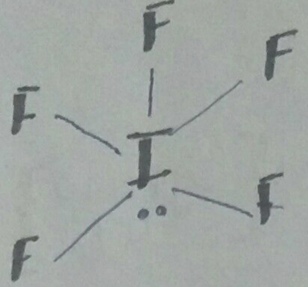
Molecular shape: Square pyramidal



iii- XeF_4

Central atom: Xenon

$_{54}Xe = [Kr]\ 4d^{10} \cdot 5s^2 \cdot 5p^6$

5s: ⇅ | 5p: ⇅ ⇅ ⇅ | 5d: □ □ □ □ □ {Ground state}

5s: ⇅ | $5p_x$ $5p_y$ $5p_z$: ⇅ ↑ ↑ | $5d_{xy}$ $5d_{yz}$: ↑ ↑ □ □ □ {Excited state}

sp^3d^2 sp^3d^2 sp^3d^2 sp^3d^2 sp^3d^2 sp^3d^2: ⇅ ⇅ ↑ ↑ ↑ ↑ {Hybridized state}

F F F F

Hybridization: sp^3d^2

Steric No: $02 + 4 = 6$

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V- ClF_5

Central Atom: Chlorine

$_{17}Cl = 1s^2 \cdot 2s^2 \cdot 2p^6 \cdot 3s^2 \cdot 3p_x^{\uparrow\downarrow} \cdot 3p_y^{\uparrow\downarrow} \cdot 3p_z^{\uparrow}$

3s	3p			3d				
↑↓	↑↓	↑↓	↑					

{ground state}

3s	3px	3py	3pz	3dxy	3dyz			
↑↓	↑	↑	↑	↑	↑			

{Excited state}

↓

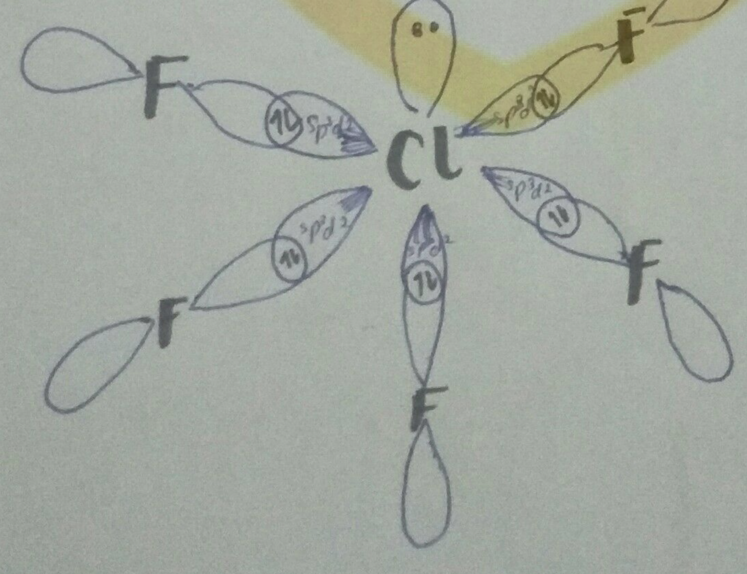
sp^3d^2	sp^3d^2	sp^3d^2	sp^3d^2	sp^3d^2	sp^3d^2
↑↓	↑	↑	↑	↑	↑
lone pair	F	F	F	F	F

{Hybridized state}

Hybridization: sp^3d^2 **Steric No** $= 1+5 = 6$

Bond pair: 05 **lone pair:** 01

Orbital Diagram: Octahedral



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Limitions of concept of Hybridization:

Some limitations of the concept of hybridization are as follows:

(i) The promotion of electrons and hybridization of orbitals are necessary for formation of hybrid orbitals. However, a promotion of electron is not necessary for the formation of hybrid orbitals. This is particularly true in multiple bonding.

(ii) It is believed that the driving force for hybridization is the energy released due to the bond formation of hybrid orbitals. We know that the carbon atom in CH_4 with ground state $2s^2 \cdot 2p^2$ does the promotion of 2s electron to 2p orbital. It can do the sp^3, sp^2 or sp hybridization of orbitals for bond formation. It depends upon the number of bonds to be formed in a particular situation. The energy consideration compel us to make sp^3 not sp^2 and sp.